# Dr. Dongbo Li

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## Professional work experience

**Postdoctoral research fellow**
(2022 – 2025)
Collaborators:
Dr. Steven Sait
Prof. Mike Fowler
Prof. Chenggui Yuan

**School of Biology, University of Leeds, UK**
Project title: *Multigenerational trophic responses to coupled short- and long-term environmental change (NERC-funded, NE/W006731/1)*.
Summary: 1) Conducted laboratory microcosm experiments to quantify eco-evolutionary dynamics in host–parasitoid systems; 2) Performed time-series analyses to assess environmental impacts on population dynamics and predator–prey oscillations; 3) Collaborated with mathematical modellers to develop experimental-informed modelling frameworks; 4) Led manuscript preparation for peer-reviewed journals and contributed to new grant proposals; 5) Coordinated laboratory activities with technical staffs and co-supervised postgraduate projects.

## Education

**PhD in *Ecology***
(2018 – 2023)
Supervisors:
Prof. Jane Memmott
Dr. Chris Clements

**School of Biological Sciences, University of Bristol, UK**
Thesis title: *The effectiveness of wildlife corridors in theory and in practice (Fully funded 4-year UoB PhD program)*.
Summary: 1) Evaluated the ecological functions of habitat corridors using integrated laboratory and field experiments; 2) Investigated how corridor-mediated dispersal regulates population and trait dynamics and predicts extinction risks; 3) Examined impacts of habitat fragmentation and landscape structure on plant–pollinator interactions.

**MRes in *Ecology***
(2015 – 2018)

**School of Biology, Nanjing University, China**
Title: *Impact of experimental warming on insect community dynamics (2018 Excellent postgraduate award)*
Summary: 1) Investigated insect community dynamics in response to warming in a field experimental system in Tibetan Plateau; 2) Conducted insect morphological identification and laboratory DNA barcoding; 3) Collected field data and contributed to manuscript preparation.

## Supervision & teaching

**2023-2024**

**Co-supervise Mbio project**
KS: Designed and optimized experimental workflow to support Mbio students; provided grouped and 1-2-1 tutorials on statistical analysis using R; reviewed and provided feedback on student reports.

**2024**

**Lead supervision of DVRS research project**
KS: Secured £3,250 in research funding to support undergraduate gaining research experiences during summer vacation; provided supervision and laboratory training related to experimental design, laboratory techniques and data analysis.

**2024-2025**

**Postgraduate mentor, University of Leeds**
KS: Organized regular meetings and discussion sessions to support a group of 5-10 postgraduates; provided guidance on academic progress, time management and research planning.

## Scientific community engagement

**Peer review activities**

**Proceedings B, The American Naturalist, Frontiers in Physiology**

**Volunteer**
2022

**BES Annual Meeting, Edinburgh**
KS: Assisted in coordinating on-site logistics to ensure the efficient delivery of workshops and parallel sessions; Provided information to support delegates; Collaborated with the organizing team to maintain event operations.

**Outreach**
2026

**Digital Research Infrastructure Retreat, Manchester**
KS: Acted as a in-person panellist to facilitate discussions on the ethics of using AI in digital research; Collaborated with UKRI and industry funders to create guidance digital research environments.

## Selected presentations

1. **Li, D.**, Coste, C. F., Yuan, C., Fowler, M. S., & Sait, S. M. The impact of environmental noise colour on the population dynamics of an insect host-parasitoid interaction. 2024 BES annual meeting, Liverpool, UK (*talk*)

2. **Li, D.,** Brough, B., Rees, J.W., Coste, C. F., Yuan, C., Fowler, M. S., & Sait, S. M. Humidity modifies heat wave effects on an insect host-parasitoid interaction. 2023 BES annual meeting, Belfast, UK (*talk*)

3. **Li, D.,** Clements, C.F. & Memmott, J. Isolation limits spring pollination in a UK fragmented landscape. 2022 BES annual meeting, Edinburgh, UK (*talk*)

## Programing and statistical skills

**Programming**: R, Julia, Stan, QGIS

**Expertise**: Time series modelling and forecasting

**Stats methods**: Spectral analysis, multivariate analysis, Bayesian ODE models, structure equation modelling, autoregressive models, spatial analysis

**Languages**: Chinese (native), English (fluent)

## Training courses

**Stats modelling**
2022

**Quantitative methods for population dynamics in R, France**
Online training course including population projection models, estimation of demographic parameters, capture-recapture models, N-mixture models.

**Theoretical ecology**
2022

**Matrix population modelling in R**
In person workshop including matrix population models, integrated projection modes and access to online database

## Publications

### ***Published***

1. **Li, D**., Clements, C. F., & Memmott, J. (2026) Linear features affect pollination success in experimental plant assemblages. *Oecologia, in press*

2. **Li, D.,** Clements, C. F., & Memmott, J. (2024). Isolation limits spring pollination in a UK fragmented landscape. *Plos one*, 19(9), e0310679.

3. **Li, D**., Brough, B., Rees, J. W., Coste, C. F., Yuan, C., Fowler, M. S., & Sait, S. M. (2024). Humidity modifies species-specific and age-dependent heat stress effects in an insect host-parasitoid interaction. *Ecology and Evolution*, 14(7), e70047.

4. **Li, D.**, Memmott, J., & Clements, C. F. (2023). Corridor quality buffers extinction under extreme droughts in experimental metapopulations. *Ecology and Evolution*, 13(6), e10166.

5. **Li, D.**, Clements, C. F., Shan, I. L., & Memmott, J. (2021). Corridor quality affects net movement, size of dispersers, and population growth in experimental microcosms. *Oecologia*, 195(2), 547-556.

6. Xi, X., **Li, D.**, Peng, Y., Eisenhauer, N., & Sun, S. (2016). Experimental warming and precipitation interactively modulate the mortality rate and timing of spring emergence of a gallmaking Tephritid fly. *Scientific Reports*, 6(1), 32284.

### ***In review***

1. Coste, C.F.D, Petersen, B., **Li, D.**, Yuan, C., Sait, S.M., Fowler, M.S., How does the rate of environmental change affect density-dependent populations? (*Ecological Monographs - in review*)

2. Liu, T., Wang L., Chen, J., Fei, M., Harvey, J.A., Xu, C., Lewis, O.T., **Li, D**., Xu, N., Xu, C., Sun S., Xi, X.. (2026) Seasonal temperature fluctuation promotes the persistence of a multi-trophic community. (*Ecology Letters- in review*)

3. Liu, T., **Li, D**., Harvey, J.A., Xi, X.. (2026) Parasitoids saved hosts by interfering a more efficient competitor. (*Ecology – in review*)

### ***In preparation (drafts upon request)***

1**. Li, D.,** Coste, C. F., Newton, J., Yuan, C., Fowler, M. S., & Sait, S. M. Environmental fluctuation and trophic interaction stabilize host-parasitoid responses to warming.

2. **Li, D**., Brown, H., Newton, J., Coste, C. F., Yuan, C., Fowler, M. S., & Sait, S. M. Warming reorganises the temporal resonant of delayed density dependence in host-parasitoid dynamics.

3. **Li, D.,** Coste, C. F., Mugabo, M., Yuan, C., Fowler, M. S., & Sait, S. M. Trophic interaction buffers the effect of environmental noise colour on host-parasitoid interaction.

4. **Li, D.,** Newton J., Rees, J.W. & Sait, S. M. Host thermal acclimation determines the stage-specific effect of heat stress on parasitoids.

5. **Li, D.,** Memmott, J & Clements, CF. Faint light arrays at night may improve pollination success in moth-pollinated plants.